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Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spatial dependency, no time dependency
 - stationary: source charge at uniform movement, steady current flow, space and time dependency
 - dynamic: source charge at fast movement, radiation, space and time dependency
 - relativistic: source charge at very speedy movement, radiation, space and time dependency
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- rest localization vs. time-dependency
 - slow vs. fast movement
 - vacuum vs. matter
 - local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
 - macro vs. micro
 - far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(c p_{0_{dipol}} + m_{0_{dipol}} \times \frac{r}{|r|} \right) \stackrel{\text{Near Field}}{\approx} \frac{1}{2} \frac{1}{2\pi} \mu_0 q \frac{v \times r}{|r|^2}, \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right) \cdot \frac{v}{c}\right]}{r^2}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Exitation:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle \mathcal{E}_m \rangle \quad (4)$$

Abbreviations

$$\Phi = \text{electric potential} \quad A = \text{magnetic potential} \quad j = \nabla \times \mathcal{H} = \frac{1}{\mu_0} \nabla \times \mathcal{B} - \nabla \times \mathcal{M} = \text{elec. current density}$$

$$\mathcal{E} = \text{Electric Field} \quad \mathcal{B} = \nabla \times A = \mu_0 (\mathcal{H} + \mathcal{M}) = \text{Magnetic Field} \quad \nabla \cdot \mathcal{B} = 0 \quad \mathcal{D} = \text{Displacement density}$$

$$\mathcal{P} = \text{Polarisation} \quad \mathcal{H} = \text{magn. exitation, magn. field strength ("magn. displacement")}$$

$$n = \text{spacial molecule density in volume}$$

$$\alpha = 4\pi r^3 = \text{constant} \quad \chi = \text{elec. susceptibility of dielectricum} \quad \kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$$

$$\langle p_m \rangle = \text{spacial average of magn. dipol moment} \quad \langle \mathcal{E}_m \rangle = \text{spacial average of electric field}$$

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant ε_0 = ...